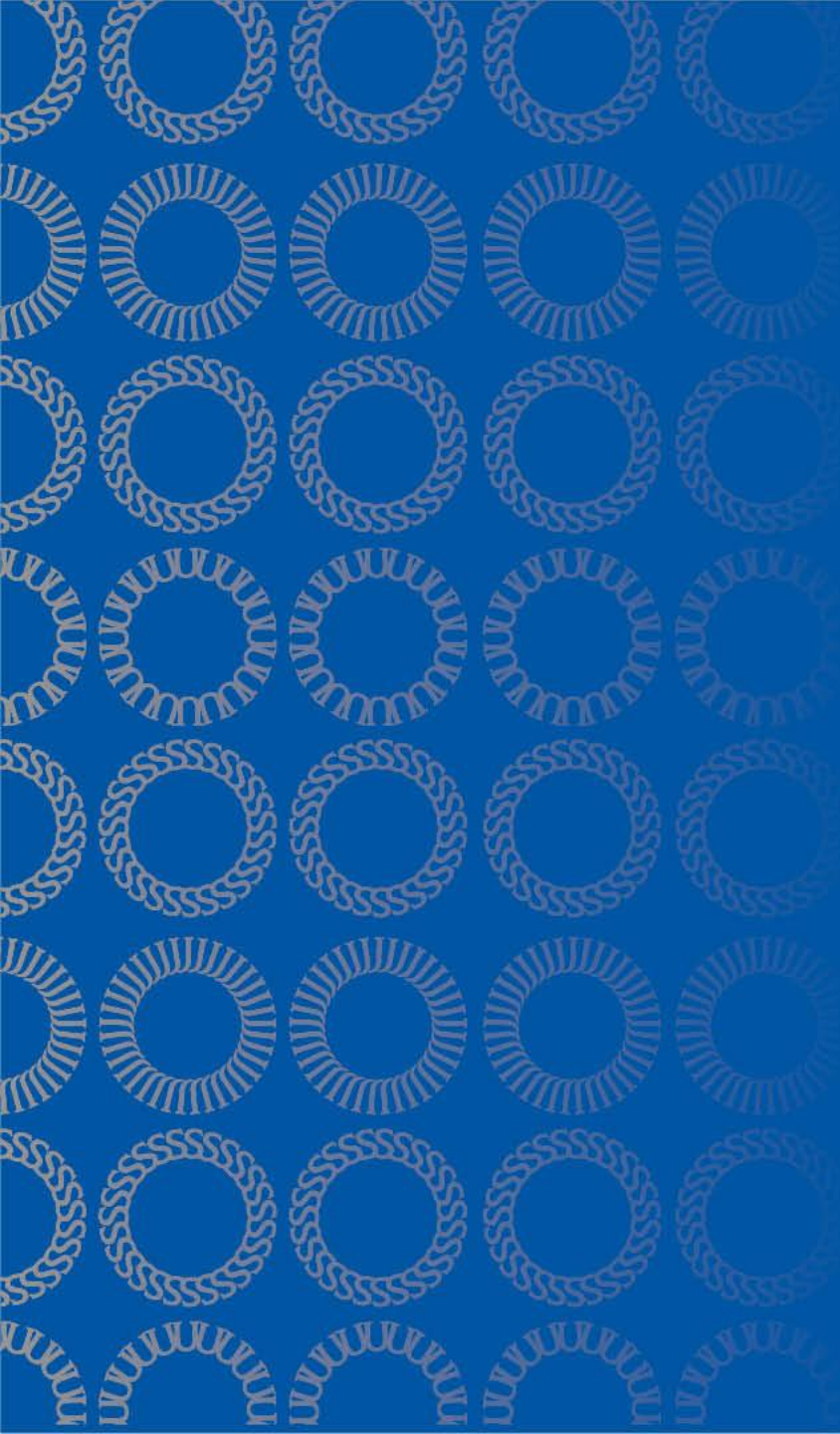




DATA 255

Deep Learning and Computer Vision in Applied Data Intelligence

Aug 19, 2026

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Instructor

Taehee Jeong

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Office hours

- By Walk-in/Appointment
- between 5pm and 6pm on Monday and Tuesday
- Or by Appointment
- Or Zoom

Instructional Student Assistant (ISA)

Mamta Jha

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Courses for Advanced Data Science

Core Courses (15 units)

- DATA 220 - Advanced Mathematical Methods for Data Intelligence 3 unit(s)
- DATA 228 - Big Data Technologies and Applications for Data Intelligence 3 unit(s)
- DATA 230 - Data Intelligence and Visualization 3 unit(s)
- DATA 245 - Machine Learning Technologies 3 unit(s) (GWAR)
- DATA 255 - Deep Learning Technologies 3 unit(s)

Elective Courses (9 units)

Complete three courses from one specialization:

Analytics Technologies

- DATA 225 - Database Systems and Management for Data Intelligence 3 unit(s)
- DATA 240 - Data Mining Techniques and Applications for Data Intelligence 3 unit(s)
- DATA 265 - Large Language Model Applications 3 unit(s)

Data Engineering

- DATA 226 - Data Warehouse and Pipeline Development 3 unit(s)
- DATA 236 - Distributed Systems for Data Engineering 3 unit(s)
- DATA 266 - Generative Models Applications 3 unit(s)

Course Description

Study deep learning theory, techniques and technologies; implement neural networks using deep learning frameworks to solve problems in various application domains; build CNNs for vision, RNNs/Transformers for sequences, and GANs for synthesis. Computer vision topics include image preprocessing and augmentation, transfer learning, object detection, segmentation with a focus on evaluation and robustness.

Prerequisite(s): DATA 230 (with a “C” or better grade).

Corequisite(s): DATA 245 .

Grading: Letter Graded.

Topics covered in this Course

- Fundamentals of Neural network
- Deep neural network (DNN)
- Convolutional neural networks (CNN)
- Autoencoder and VAE
- Generative adversarial networks (GAN)
- Diffusion models

Course Protocol

- Punctuality:
 - Arrive on time.
- Attendance Responsibility:
 - If you miss a class, it is your responsibility to catch up on any announcements, changes to deadlines, or materials covered.
- Device Etiquette:
 - Turn off or mute your cell phone during class.
- Classroom Conduct:
 - Refrain from engaging in activities that are irrelevant or distracting to others. No food is allowed.
- Active Participation:
 - Stay engaged and contribute to discussions and activities to enhance the learning experience for everyone.

Course Protocol

- Students Are Not Allowed to Record

Students are prohibited from recording class activities (including class lectures, office hours, advising sessions, etc.), distributing class recordings, or posting class recordings.

Materials created by the instructor for the course (syllabi, lectures and lecture notes, presentations, etc.) are copyrighted by the instructor.

This university policy (S12-7) is in place to protect the privacy of students in the course, as well as to maintain academic integrity by reducing the instances of cheating. Students who record, distribute, or post these materials will be referred to the Student Conduct and Ethical Development office. Unauthorized recording may violate university and state law.

It is the responsibility of students who require special accommodations or assistive technology due to a disability to notify the Accessible Education Center(AEC) at <http://www.sjsu.edu/aec>. AEC will in turn contact the instructor as needed.

Textbook

This class does not have a required textbook. Lecture slides and selected readings from online materials will cover most of the topics.

Reference textbooks (not required)

- Math and Architectures of Deep Learning, Krishnendu Chaudhury
- Hands-On Machine Learning with Scikit-Learn & TensorFlow, Aurelien Geron
- Deep Learning with Python, Francois Chollet
- Neural Networks and Deep Learning, Charu C. Aggarwal
- Introduction to Deep Learning, S. Skansi
- Deep Learning for Vision Systems, Mohamed Elgendy
- Generative Deep Learning(2nd Ed), David Foster
- GANs in action, Jakub Langr & Vladimir Bok
- Hands-On Generative AI with Transformers and Diffusion Models, Omar Sanseviero et al.
- Build a Large Language Model (from Scratch), Sebastian Raschka

Course Prerequisites

- Proficiency in Python
 - The majority of class assignments will be in Python. You should be familiar with Jupyter notebook and PyTorch.
- College Calculus, Linear Algebra
 - You should be comfortable understanding **Chain Rule, Partial Derivatives, Matrix Multiplication, and Matrix Transposition**
- Basic Probability and Statistics
 - You should know the basics of probabilities, Gaussian distributions, mean, standard deviation, etc.
- Machine Learning
 - You should be comfortable with the basics of machine learning

Academic integrity: Following is prohibited

- Copying from another's work or allowing someone to copy from your own work
- Representing someone else's work as your own
- Plagiarism

Copying and other forms of cheating will not be tolerated and will result in a zero score for the quiz, assignment, exam (minimal penalty) or a failing grade for the course, possibly combined with other disciplinary actions from the university.

1.2 *PLAGIARISM*

San José State University defines plagiarism as the act of representing the work of another as one's own without giving appropriate credit, regardless of how that work was obtained, and submitting it to fulfill academic requirements.

Plagiarism includes:

- 1.2.1 Knowingly or unknowingly incorporating the ideas, words, sentences, paragraphs, or parts of sentences or paragraphs, or the specific substance of another's work without giving appropriate credit, and representing the product as one's own work;
- 1.2.2 Representing another's artistic or scholarly works, such as computer programs, instrument printouts, inventions, musical compositions, photographs, paintings, drawings, sculptures, novels, short stories, poems, screen plays, or television scripts, as one's own.

AI tools

- Prompting LLMs such as ChatGPT is permitted for low-level programming questions or high-level conceptual questions about language models but **using it directly to solve the problem is prohibited.**
- **AI-generated solutions (for example, Assignments, Project report, Project presentation, Lab presentation) will not be accepted and will be considered cheating.**
- If AI use is necessary for any assignment, It will be announced when and how it can be used.

In-class quizzes (10%)

- Students will be given in-class quizzes that will test their understanding and retention of presented materials.
- Students are required to submit their own work and cannot submit work for others.
- The lowest score of the quizzes (only one) would be dropped and not counted.
- *Students who submit in-class activity assignment including survey, but were not in class, as well as the students who submit on someone else's behalf are violating the academic integrity policy and will be reported immediately to the office of Student Conduct and Ethical Development.*

Assignments (10%)

- All assignments are needed to submit via Canvas by the deadline.
- If anything goes wrong, please contact ISA or the instructor. It is a student's responsibility to figure out if the instruction of assignments is not clear for him/her.
- You can submit as many times as you'd like until the deadline; we will grade only the last submission.
- The lowest score of the Assignments (Only one) would be dropped and not counted.
- All assignments might be re-evaluated at any time during the semester.

Lab(20%)

- All Lab assignments are based on Mobilint NPU.
- All Lab assignments are needed to submit via Canvas by the deadline.
- If anything goes wrong, please contact ISA/RA or the instructor. It is a student's responsibility to figure out if the instruction of assignments is not clear for him/her.
- Class engagement bonus might be applied for Lab presentation (TBD).

Late days for Assignments/Labs

- Students are allocated a total of **6** late days, which can be used for either assignments or labs.
- One late day extends the deadline by 24 hours.
- Students can use up to three late days per assignment/lab. You will receive no credit for a submission which is more than three days late.

Group project (20%)

- Groups of 4 - 5 students will be formed to work on a semester-long project.
- The project should be based on the course materials covered in class. Some points would be deducted for any content that is not covered in class.
- Each group member is expected to participate in every phase of the project.
- Class engagement bonus might be applied for Group project presentation (TBD).
- The project will be graded based on the following components:
 - a) progress check-up: 5 pts
 - b) final presentation: 10 pts
 - c) technical report: 5 pts

Group project schedule

tentative and subject to change

- Aug 26: Group formation (on canvas)
- Sept 9: Project proposal (within 5 mins)
- Nov 4: Project progress checkup (within 7 mins) – 5 pts
- Dec 2: Submit presentation slides/ Project presentation (within 15 mins) – 10 pts
- Dec 2: Submit source codes and report (within 6 pages)– 5 pts

Mid-term Exam (15%)

- Exam will be based on the course material covered in class.
- Wednesday, Sept 30th (tentative and subject to change)
- 6:00pm to 8:00pm

Final Exam (25%)

- This course has a comprehensive final exam.
- Exam will be based on the course material covered in class.
- Wednesday, Dec 9th (The date is final.)
- 5:30pm to 7:30pm

Determination of Grades

In-class quizzes	10 %
Assignments	10 %
Lab	20 %
Group Project	20 %
Mid-term exam	15 %
Final exam	25 %

Grading criteria of quizzes, Assignments, Labs, Mid-term exam, Final exam would be different.

Grading breakdown

The final grades will be calculated based on the following:

- (A+) $\geq 96\%$,
- (A) $\geq 93\%$ and $<96\%$
- (A-) $\geq 90\%$ and $<93\%$
- (B+) $\geq 86\%$ and $<90\%$
- (B) $\geq 83\%$ and $<86\%$
- (B-) $\geq 80\%$ and $<83\%$
- (C+) $\geq 76\%$ and $<80\%$
- (C) $\geq 73\%$ and $<76\%$
- (C-) $\geq 70\%$ and $<73\%$
- (D) $\geq 60\%$ and $<70\%$
- (F) $< 60\%$

Curve grading is applied:

Final grade would be adjusted based on the overall performance of the class, rather than using a fixed grading scale.

Grading breakdown

The final grades will be calculated based on the following:

(A+) $\geq 93\%$,

(A) $\geq 90\%$ and $< 93\%$

(A-) $\geq 86\%$ and $< 90\%$

(B+) $\geq 83\%$ and $< 86\%$

(B) $\geq 80\%$ and $< 83\%$

(B-) $\geq 76\%$ and $< 80\%$

(C+) $\geq 73\%$ and $< 76\%$

(C) $\geq 70\%$ and $< 73\%$

Regrade requests

- If you believe that the course staff made an objective error in grading in (Quiz, Assignment, mid-term exam, final exam), you may request to regrade **within 3 days** after the grades are released.

Schedule

Go to <https://sjsu.instructure.com/>

Find the course/SJSU Official Syllabus

Inference chips

- Amazon: Trainium3, Inferentia
- Google: TPU v10
- Microsoft: Maia 200
- Meta: MTIA
- Telsa: AI8
- Celarbas

NPU: Mobilint

- We would like to thank mobilint¹ for sponsoring for this class



MLA100 Low Profile
NPU PCIe Card

ARIES Specifications

Performance

80 TOPS

Memory Bandwidth

66.7 GB/s

AI Performance

ResNet-50: 3,082 FPS

Host Interface

PCIe Gen 4.0 8-lane

Power (TDP)

25 W

AI Performance

YOLOv9-S: 627 FPS

Memory Capacity

16 GB (Optional 32 GB) LPDDR4, 4X

AI Performance

MobileNetV2: 11,551 FPS

AI Performance

YOLOv9-C: 254 FPS

¹<https://www.mobilint.com/>

Lab1: Image classification

1. Build a model in PyTorch
2. Train the model
3. Convert to ONNX
4. Quantize/compile ONNX to MXQ
5. Deploy MXQ on NPU



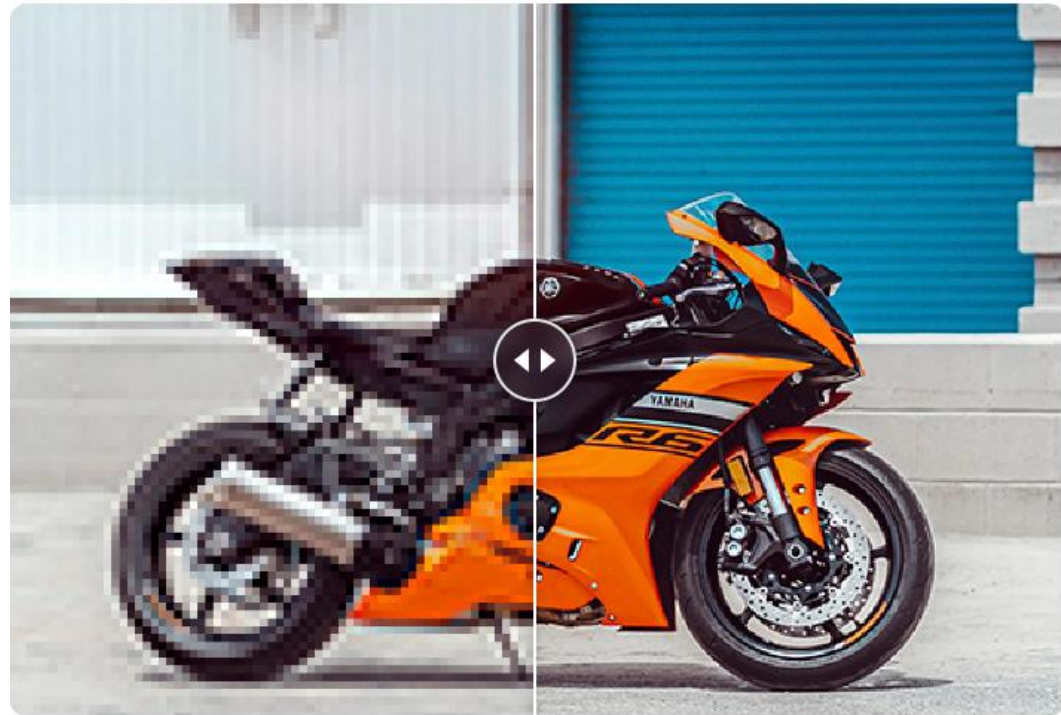
Input: x
Image pixels



Output: y
Object in image

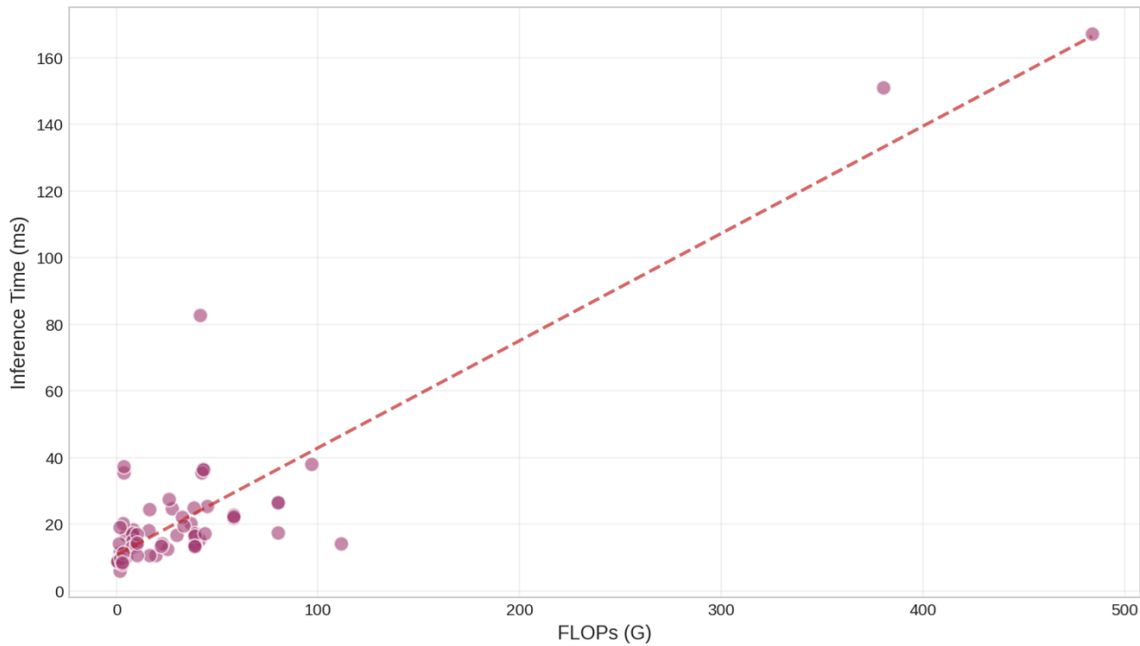
Lab2: Super resolution (image restoration)

- Super Resolution uses machine learning to clarify, sharpen, and upscale the photo without losing its content and defining characteristics.
- Applications:
 - Photography
 - Surveillance
 - Medical diagnostics
 - remote sensing
 - video gaming using consoles
 - Augmented Reality (AR)/Virtual Reality (VR)/Mixed Reality (MR) using mobile devices

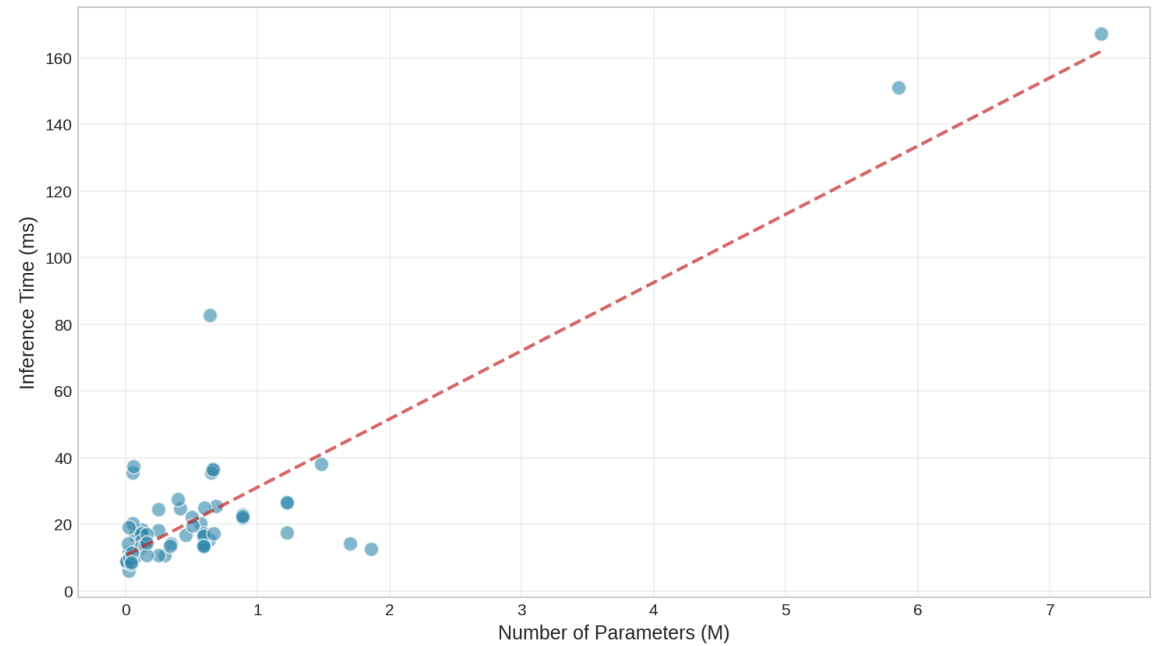


Lab3: Optimizing SR model to reduce latency

Inference Time vs FLOPs



Inference Time vs Parameters



Group Project idea¹

NPU typically has a limited hardware resource such as memory and computing power.

Labs are designed to perform [256x256x3] pixel to [256x256x3] pixel.

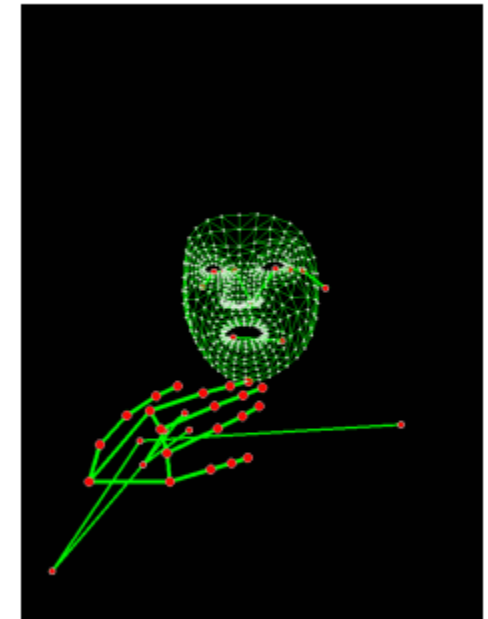
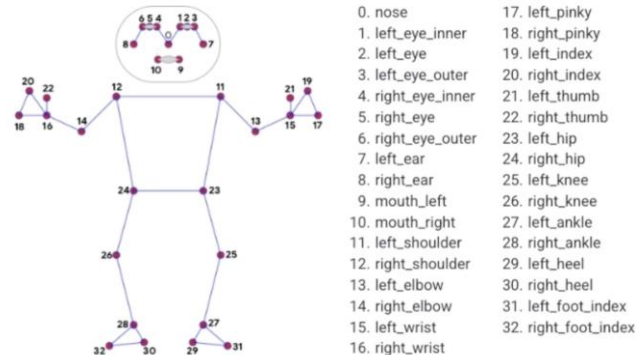
Instead, your task is to reconstruct sharp and clear images from degraded or blur or noised inputs and **upscale** it on NPU.

- Input: [256x256x3]
- Output: [1024x1024x3]
- On *Mobilint* NPU

Group Project idea²

Extract Key points from video

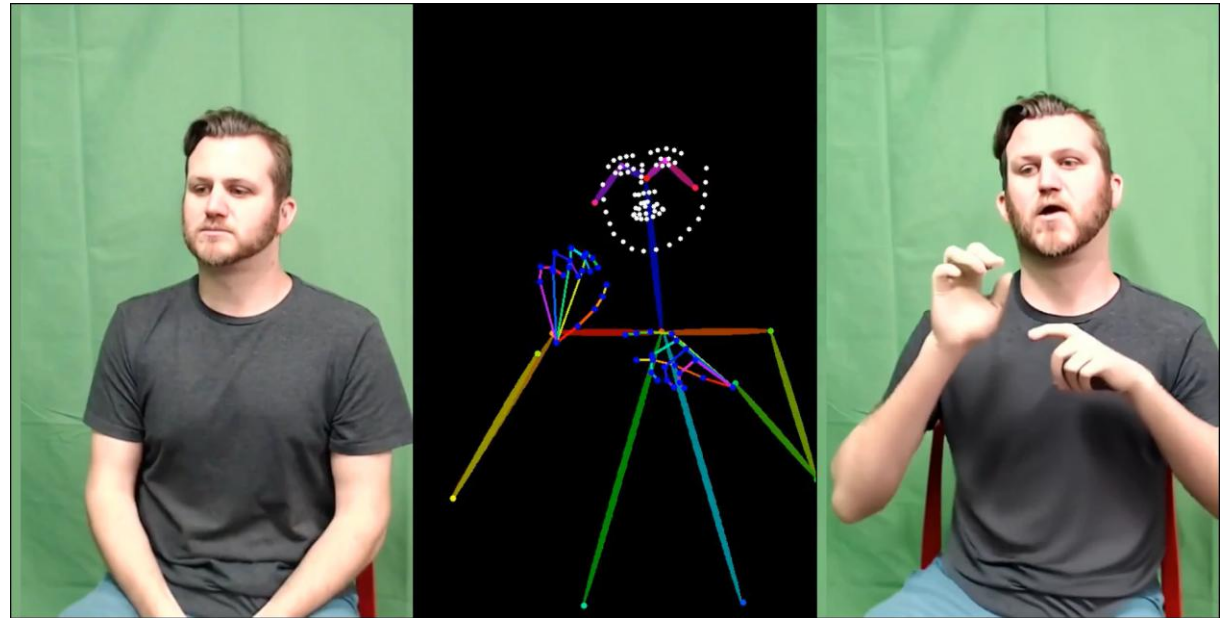
- Mouth movement
- Hand gestures
- Instead of relying on any package, build it from scratch



Group Project idea³

Generate video (movie clip)

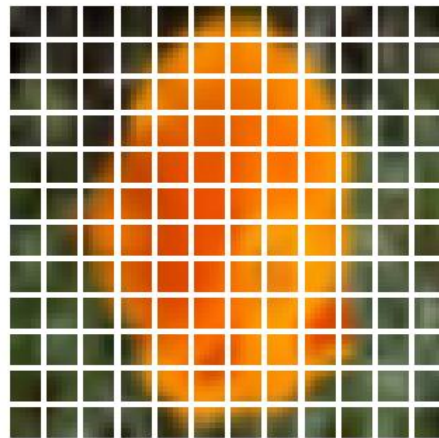
- Given an image and key points
- Generate signing video
- Instead of relying on any package, build it from scratch



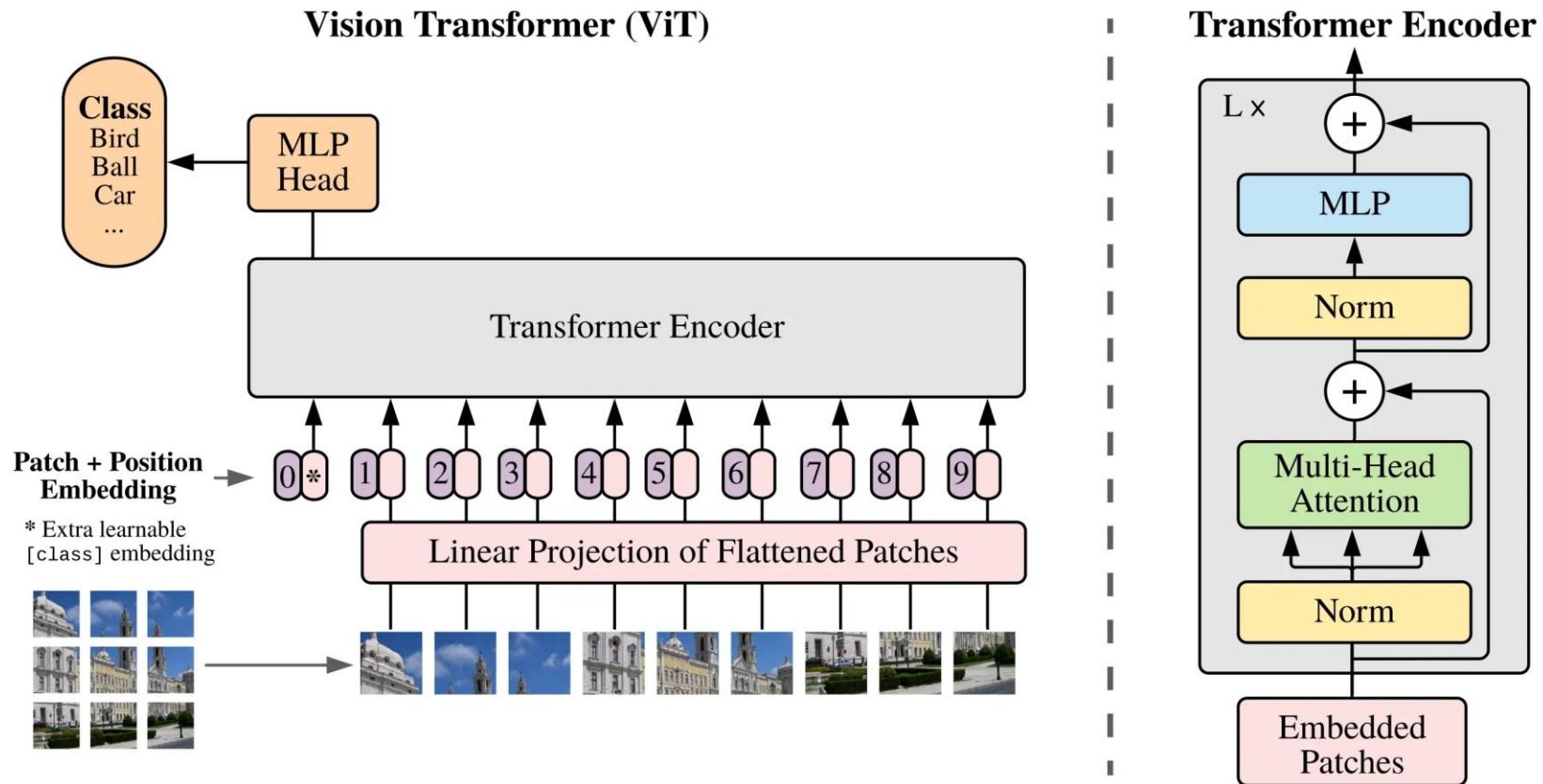
Group Project idea⁴

Image classification with Vision Transformer

- Instead of relying on any model, build it using PyTorch from scratch



Group Project idea⁴



Group Project idea⁴

Architecture of Vision Transformer

1. Patch Embedding
2. Position Embedding
3. Class Token
4. Transformer Encoder
5. MLP Head

Group Project idea⁵

YOLO (Object detection)

- Instead of relying on any model, build it using PyTorch from scratch

Era / Generation	Key Features & Modules	Structural Shift
Early Era (v1 - v3)	GoogleNet-based, Darknet-53 backbone, Feature Pyramid Networks (FPN)	Introduced the initial anchor-based, single-stage pipeline.
Mid Era (v4 - v7)	CSPDarknet, SPPF layer, PANet neck fusion	Maximized inference speeds through specialized feature-pooling and cross-stage paths.
Modern Era (v8 - v12)	C2f blocks, Decoupled Heads, Attention mechanisms	Shifted to Anchor-Free design with independent classification and regression branches.
Next-Gen (YOLO26+)	C3k2, SPPF residuals, NMS-free inference	Native end-to-end detection that removes post-processing logic for strict edge efficiency.

Group Project idea⁶

Agentic workflow

Build a function for object detection

Integrate with LLM (agentic workflow)

Apply a use-case, for example, counting crops or human in an image or video

Group Project idea⁷

Diffusion model

- Instead of relying on any model, build it using PyTorch from scratch

Core Components

- Forward Process (Diffusion)
- Reverse Process (Denoising)
- Timestep Conditioning
- Conditioning Guidance (Text embedding)

Criteria for Group project: progress checkup

Criterion	Description	Points
1. Project Domain / Problem	Clearly explain the area or domain your project belongs to (e.g., medical imaging, scene understanding, anomaly detection, etc.). Describe the problems /challenges in this domain you are targeting.	1.0
2. Current Baseline	Describe the existing works (what baseline deep learning models exist). What is the current status or limitations of existing works, and what gaps are still remained.	1.0
3. Dataset and Evaluation Metrics	Identify the relevant dataset(s) used in prior works and specify the evaluation metric(s) (e.g., accuracy, FID) and baseline performance.	1.0
4. Your Approach / Technical Novelty	Explain your approach in terms of technical novelty or architectural improvement compared to the baselines (existing works from your literature survey), and how it aims to overcome the identified limitations.	1.0
5. Presentation	• Does the presentation follow logical flow? • Is the presentation to attract audience? • Is the presentation to deliver content effectively? • Is the presentation within the time limit?	1.0

Criteria for Group project: final presentation

Criterion	Description	Points
1. Project Domain / Problem	Clearly explain the area or domain your project belongs to (e.g., medical imaging, scene understanding, anomaly detection, etc.). Describe the problems /challenges in this domain you are targeting.	2.0
2. Current Baseline / Dataset and Evaluation Metrics	Describe the existing works (what baseline deep learning models exist). What is the current status or limitations of existing works, and what gaps are still remained. Identify the relevant dataset(s) used in prior works and specify the evaluation metric(s) (e.g., accuracy, FID) and baseline performance.	2.0
3. Your Approach / Technical Novelty	Explain your approach in terms of technical novelty or architectural improvement compared to the baselines (existing works from your literature survey), and how it aims to overcome the identified limitations.	2.0
4. Results	What are the performance of your approach? Compare it with the existing works (baselines) using the evaluation metrics.	2.0
5. Presentation	• Does the presentation follow logical flow? • Is the presentation to attract audience? • Is the presentation to deliver content effectively? • Is the presentation within the time limit?	2.0

Technical report for Group project

- Six-page (for double column) or ten-page (for single column) not counting references
- It should include:
 - Title
 - Authors
 - Abstract
 - Introduction: Description of purpose, motivation
 - Related works (optional)
 - Method(s)
 - Results: Supporting images/tables/figures
 - Conclusions

Criteria for Group project: report

Criterion	Description	Points
1. Project Domain / Problem	Clearly explain the area or domain your project belongs to (e.g., medical imaging, scene understanding, anomaly detection, etc.). Describe the problems /challenges in this domain you are targeting.	1.0
2. Current Baseline /Dataset and Evaluation Metrics	Describe the existing works (what baseline deep learning models exist). What is the current status or limitations of existing works, and what gaps are still remained. Identify the relevant dataset(s) used in prior works and specify the evaluation metric(s) (e.g., accuracy, FID) and baseline performance.	1.0
3. Your Approach / Technical Novelty	Explain your approach in terms of technical novelty or architectural improvement compared to the baselines (existing works from your literature survey), and how it aims to overcome the identified limitations.	1.0
4. Results	What are the performance of your approach? Compare it with the existing works (baselines) using the evaluation metrics.	1.0
5. Writing skill	its coherence (All points should connect logically to the main topic with clear transitions)	1.0